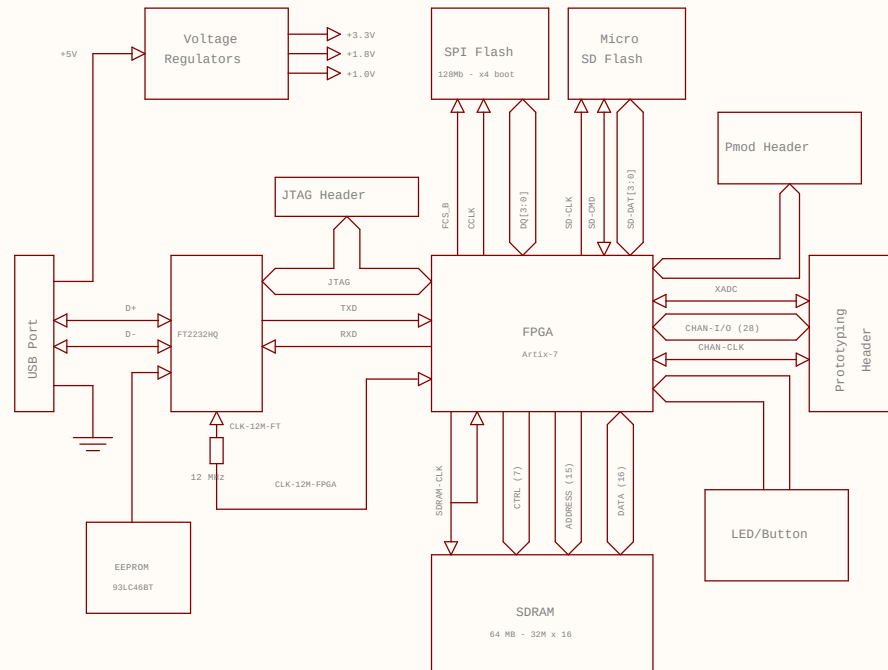
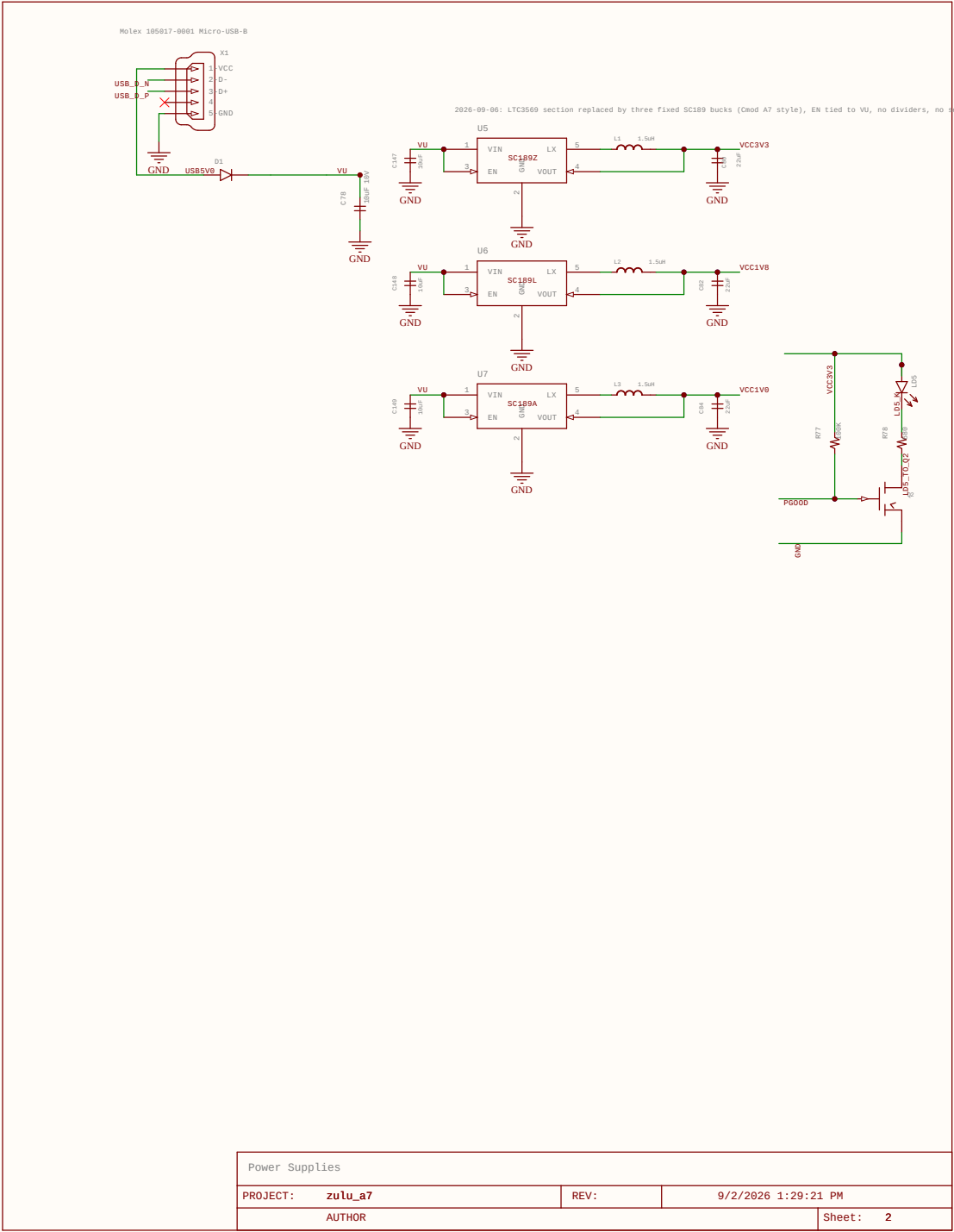


SPI flash sits on the FPGA's dedicated master-SPI config pins: CCLK C11, FCS_B K19, DQ[3:0] on D18/D19/G18/F18. Read at power-up.

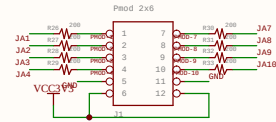
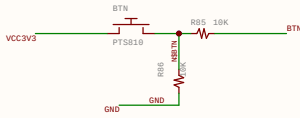
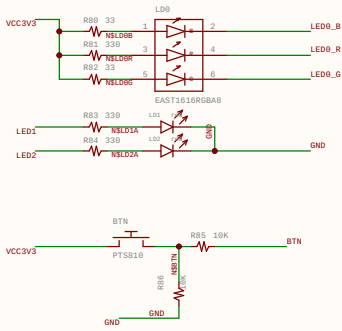
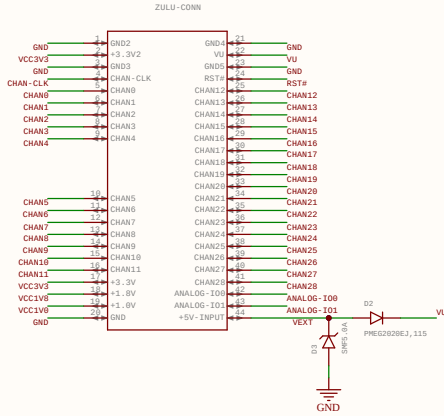
microSD runs the native 4-bit SD bus, not SPI: CLK, CMD and DAT[3:0] on six bank-34 balls of its own, so a card cannot touch the config bus.



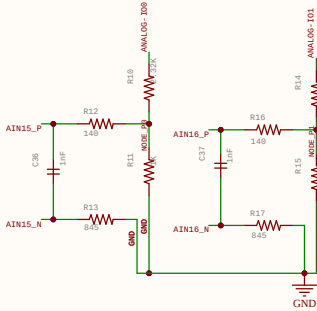
Block Diagram		
PROJECT: zulu_a7	REV:	9/2/2026 1:29:21 PM
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X2: 44 pins on a 2.54 mm grid, rows 0.700 in apart, board 69.85 x 20.32 mm.
Numbered right to left and straight through: top row 1-20, bottom row 21-44.
TOP ROW, x 50.69 down to 1.27: GND, +3.3V, GND, then CHAN-CLK and CHAN0..CHAN3
(pins 4-8), CHAN4 (pin 9), CHAN5..CHAN11 (10-16), +3.3V, +1.8V, +1.0V, GND.
FOUR positions between pin 9 and pin 10 -- x 26.82, 34.29, 31.75, 29.21 -- are
RESERVED, not spare: the landing for the edge-mounted micro-USB X1, 7.80 mm of
copper in 11.176 mm clear, centred with 1.69 mm each side.
BOTTOM ROW, contiguous, x 50.69 down to 1.27: GND, VU, GND, RST#, CHAN12..CHAN28
(pins 25-41), ANALOG-IO0, ANALOG-IO1, +5V-INPUT.
FIVE grounds: pins 1, 9, 20, 21, 23. Pin 22-VU SOURCES 5 V, pin 44 only takes it.

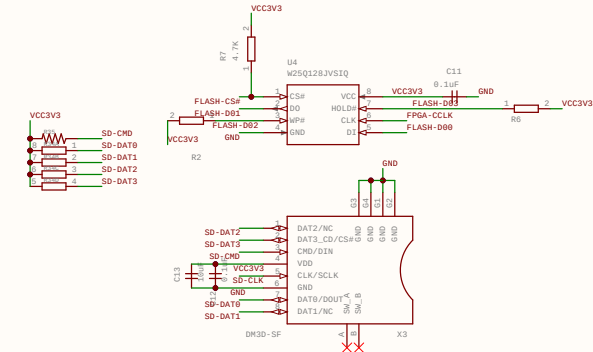


Digilent 12-pin Pmod (JA). Pins 1-4 and 7-10 are signals through 200R series resistors; pins 5/11 are GND and 6/12 are VCC3V3 -- so a 2x6 header carries EIGHT signals, not twelve, which is why the names skip JA5/JA6. No decoupling here: the pmod spec puts bypassing on the module, and VCC3V3 already carries ~170uF. The Cmod A7 has none at its Pmod either. JA1-JA4 (the top row, the half a Pmod uses for SPI) are all in bank 16: A14 A15 B15 A16. JA7 is C15, also bank 16. JA8/JA9/JA10 are W16/W2/W7 -- banks 14/35/34 -- because nothing else was free. All five banks are VCC0=3.3V, so this costs timing coherence, not levels.



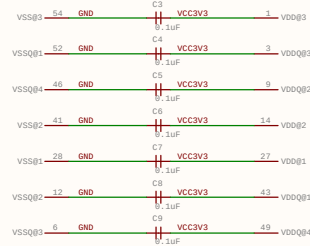
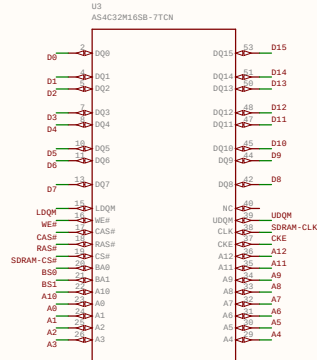
General IO			
PROJECT:	zulu_a7	REV:	9/2/2026 1:29:21 PM
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MASTER SPI BOOT (x4). R2/R1/R0 - 0:0:1 -- R2B pulls WB to VCC3V3, R21/R22 pull M1/M2 to GND -- so at power-up the FPGA drives CLK itself and clocks its bitstream out of U4 with no USB and no host present. That mode can only use the dedicated configuration pins: CLK C11, FCS_B K19, D00_M0S1 D18, D00_DIN D19, D02 G18, D03 F18. No other ball will do, which is why SDRAM A6 had to move off K19 (it is now on D17, still bank 14). After DONE the six pins revert to ordinary user I/O. R2A/R2B keep WP#/HOLD# high through the x1 command phase and through the float window before the FPGA drives D02/D03 (PUDC_B is tied high, so the FPGA has no internal pull-ups). R2D keeps FCS_B deselected. Vivado: set_property CONFIG_MODE SPIx4 [current_design] set_property BITSTREAM.CONFIG SPI_BUSWIDTH 4 [current_design] The microSD is deliberately NOT on this bus any more: an inserted card sits in native SD mode watching bitstream traffic go past on what it believes is CMD, and one 48-bit window with a valid CRC7 is enough to make it answer and drive the line the FPGA is reading. X3 now has its own CLK/CMD/DAT0 on R2/T1/T2, the bank-34 balls the flash just vacated, with R9 damping DAT0 where the socket stub is.



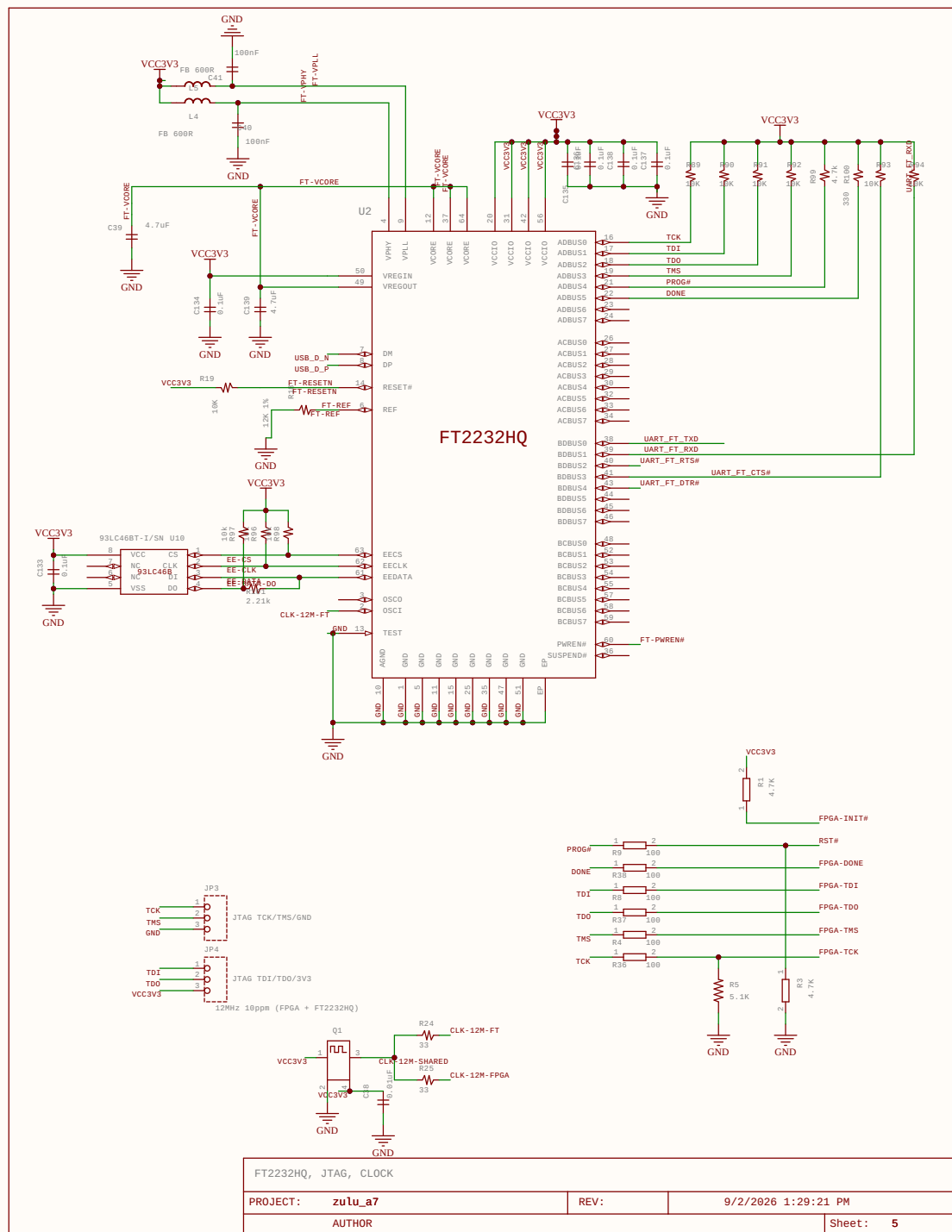
microSD runs the NATIVE 4-BIT SD BUS, not SPI. CLK, CMD and DAT0-DAT3 -- six signals, all in bank 34 byte group T0 (R2 T1 T2 U1 V2 W2), so they can be constrained as one bus. DAT1/DAT2 came from prototyping-header CHAN1/CHAN0; DAT3 is the old SPI chip-select, which needs no new pin. R34/R35 are the 4.7k pull-ups the SD spec requires on CMD and DAT0-DAT3. R9 (100R on DAT0 only) was removed: one damped line out of four is a skew, not termination. 2026-09-06: X3 is now the Hirose DM30-SF push-pull socket (HRS 600-08025-8), DM3 catalog p9. Same 8-contact pinout; shell pads G1-G4 to GND; card-detect switch A/B (closed with card) left open. Needs its own footprint on the PCB.

U3 = AS4C32M16SB-7TCN: 512Mbit (32M x 16) = 64MB. Pin-for-pin and dimensionally identical to the 250MB AS4C32M16SA-07T0 it replaces -- all 54 pins and every TSOP-II dimension checked against both datasheets, and pin 48 is NC on both. ONE DIFFERENCE, AND IT IS NOT A PIN: the column address is A0-A9 (1024 columns), not A0-A8 (512). The memory controller must be set to 3D column bits, or it reaches only half the array. Row A0-A12, banks BA0/BA1 and refresh (8192 cycles / 64ms) are unchanged. -7TCN since 2026-09-08 (-07N err); 343 M2 max at C13, tRCO/1RP 21, tRC 63 ns, TA 0-70 C, IDD1 max 110 mA.

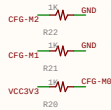


U3 DECOUPLING -- each cap bridges one supply pin to the ground pin OPPOSITE it. In the 54-pin TSOP-II, pin n faces pin 55-n, and every supply pin faces a ground of its own domain: VDD 1/24/27 to VSS 54/41/28, VDDQ 30/9/43/49 to VSSQ 52/46/12/6. Alliance isolates VDDQ from VDD for DQ noise immunity -- LAYOUT MUST KEEP THE PAIRS: each cap on the underside spanning its two pins, own vias, VDDQ returning to VSSQ and VDD to VSS. C18 is bulk and pairs with nothing. Do NOT split the ground plane.

Memory			
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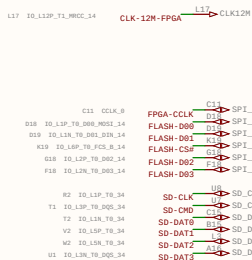


PROTOTYPING HEADER



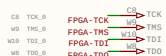
CLK

FLASH

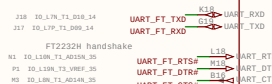


JTAG

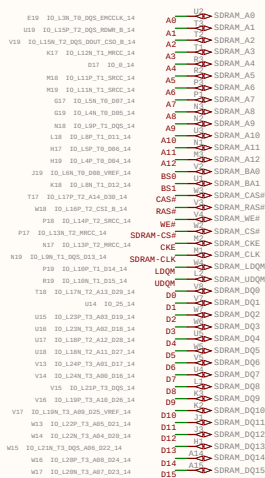
ANALOG



FT2232H



SDRAM



UNCONNECTED BALLS
XADC analog (unused)

Thermal diode (unused)

Battery backup (unused)

TP - float (UG482 T5-4)

B2 MGITXR1 216

```

88  MCTRES CLKOP_218
89  MCTRES CLKOP_219

```

B10 MGTREFCLK1P_210

A2 MGTPTXN1_216

Pmod JA (bank 16 x5)

A15 ID_LEN_TO_VREF_16

A16 IO_L12P_T1_MRCC_16

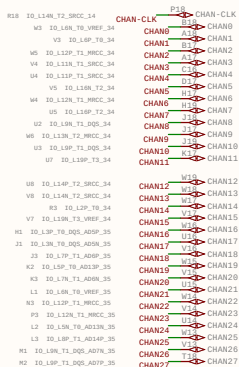
W12 IO L16N T2 A15 Q31 14

N3 10.1320.T3.MRCC.34

100 50 0 -50 -100

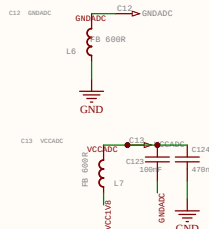
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	0.0000
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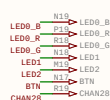


The six pins above CCLK₁ are the ONLY halves master-SPI configuration can use (UCG470). They are the ordinary user I/O once DONE goes high. SDRAM A4 master-SPI → D17 to free FCS_B; the ESP reset moved D17 → B15. microSD is on its own CLK/CMD/DAT0 so a card cannot fight the configuration bus.

The six microSD signals below CCLK/Flash are the native 4-bit SD bus: CLK CMD DAT0-DAT3. All sit in bank 34 byte group 10 (S2 I17 to U3 V2 W0). DAT3/DAT2 look CHAN1/CHAN0 off the prototyping header; DAT3 is the SD SPI chip-select, rendered, only two pins were spent.



LED / BUTTON



FPGA Connections			
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